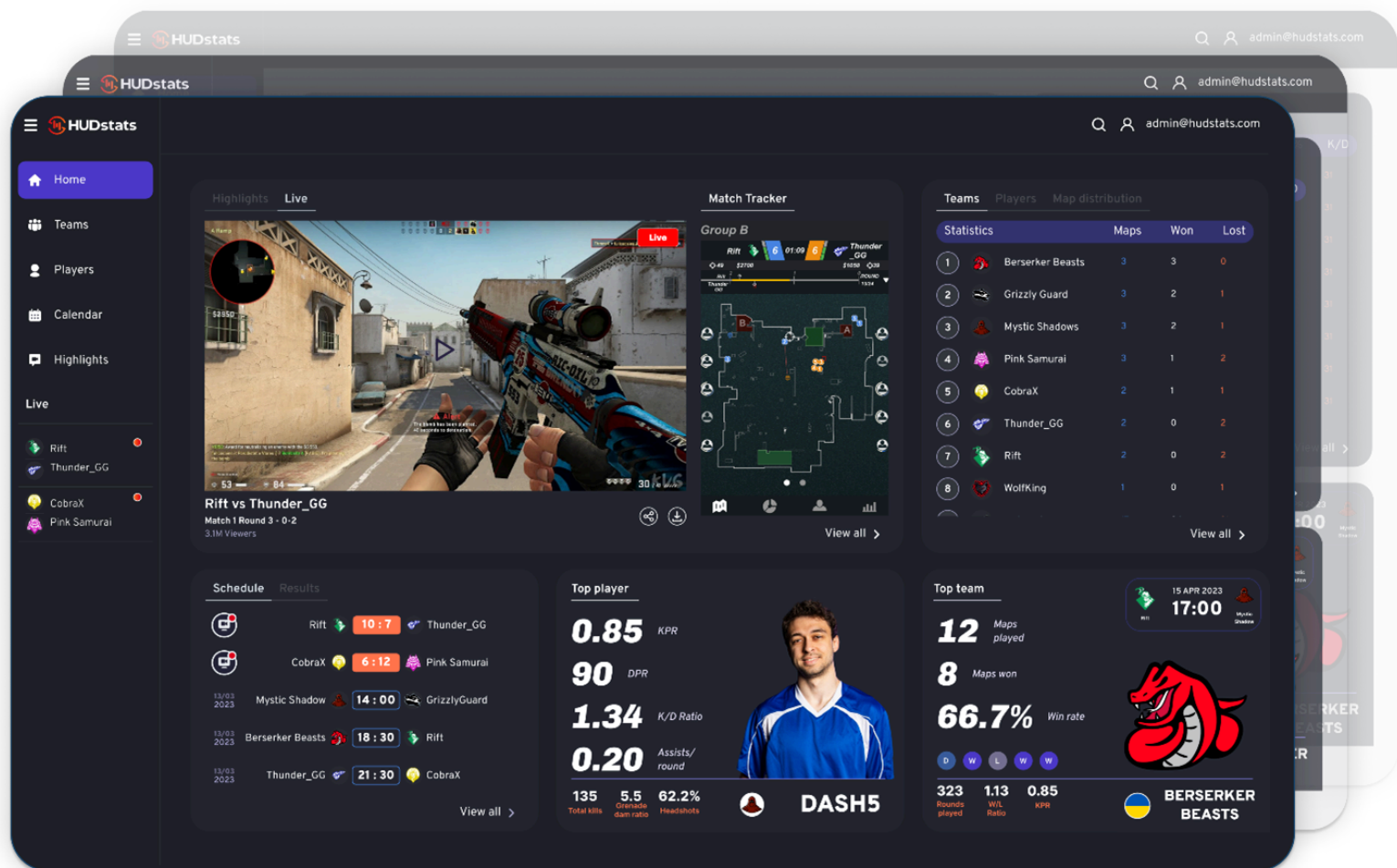


HUDstats switches from OpenAI to Amazon Bedrock to advance esports storytelling using generative AI

by Emily McKinzie | on 08 APR 2024 | in [Amazon Bedrock](#), [Game Development](#), [Generative AI](#) | [Permalink](#) | [Share](#)



In the past decade, esports has evolved into a billion-dollar industry, with dozens of leagues, thousands of athletes, teams, franchises, and rising fan culture. In this realm, as in the world of physical sports, storytelling is key to success.

Turning pixels into stories is the focus for [HUDstats](#), a Netherlands/Hague-based esports data and analytics company on the forefront of benchmarking innovation in the global esports market. With its Advanced Video Analysis (AVA) artificial intelligence (AI) and Machine Learning (ML) solution, HUDstats extracts data in real-time from live esports broadcasts using deep learning or from the games API; data is then automatically aggregated into statistics, and fed into a user-friendly dashboard, providing accessible content for entertainment and coaching purposes, or accurate real-time data for betting purposes.

"When we started the company in 2020, our ideas revolved around building positional player tracking and data visualization solutions for traditional sports, but when they were paused due to the pandemic, we pivoted to esports, and started with data widgets using a similar approach. Moving from there to building a whole AI-driven data infrastructure and generative AI storytelling solution for the esports industry felt pretty natural. It was a big step forward, or maybe a couple of steps, but it all came together naturally," said HUDstats CEO Andrei Balanescu.

To drive innovations in this fast-moving market, HUDstats prioritizes reliable, battle-tested resources that help it move quickly and prototype tools their clients haven't yet imagined, which is why the company adopted [Amazon Bedrock](#) for generative AI.

Gaining flexibility with Amazon Bedrock

AI is central to HUDstats' core application, but as its business and client opportunities grew, HUDstats needed a solution that enabled flexibility and scalability in addition to raw power. HUDstats is built on Amazon Web Services (AWS), but initially leveraged OpenAI for generative AI workloads. Bumping into limitations with the models available, the team began to explore Amazon Bedrock, a fully managed service that offers a choice of high-performing foundation models from leading AI companies like Anthropic, Cohere, Meta, Stability AI, and Amazon via a single API, along with a broad set of capabilities for building generative AI applications with security, privacy, and responsible use of data. HUDStats found that more choice of accessible foundation models means better business outcomes, leading it to make the switch.

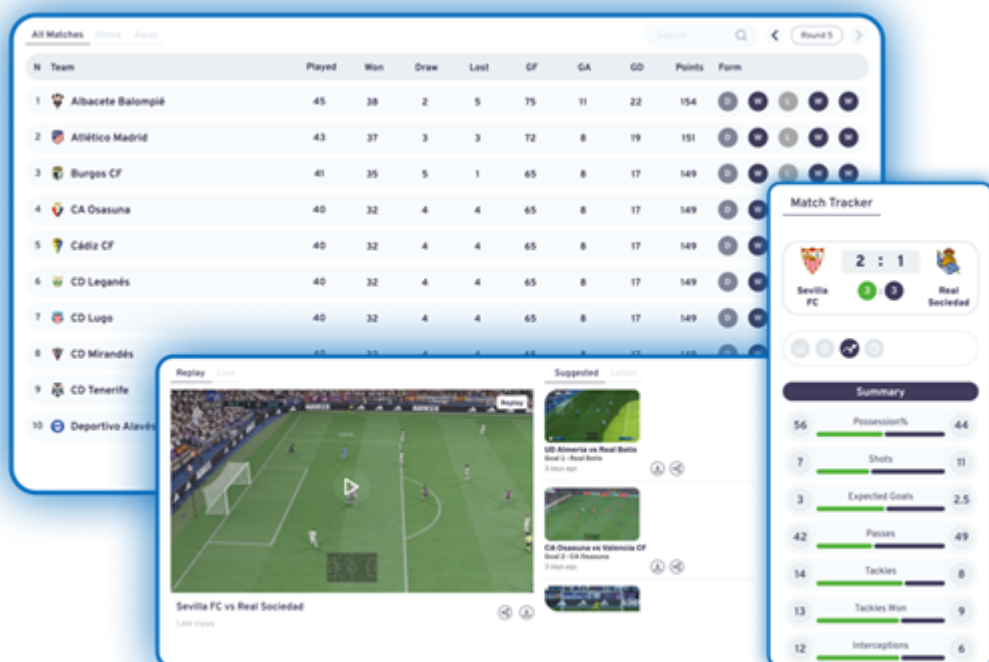
Balanescu said, "We can fine-tune our foundation models in Amazon Bedrock so they align with our needs. Having more options makes our life easier in how we validate and iterate new ideas."

CTO Hristo Yankov elaborated, "Switching to Amazon Bedrock really changed the game for us. Unlike with OpenAI, we got this incredible flexibility and scalability. It lets us tweak settings, experiment with various models, and tailor everything to our specific needs and data. Honestly, our experience has been fantastic and we're looking forward to what the future holds in terms of Amazon Bedrock. We've got plenty of other ideas in the box."

Turning data into storytelling for LaLiga

One of the most exciting applications of HUDstats' innovation is its solution for FC24 official esports league [LaLiga](#). The league uses HUDstats' AI-driven storytelling dashboard to automate esports data collection, enrich narratives, and scale content production. Integrating HUDstats' AI data collection and aggregation technology, paired with rich content visualizations and video highlights functionality, the storytelling dashboard allows LaLiga to extract more value from its esports data. It also eliminates the need for manual processing and aggregation of game data.

LaLiga seamlessly integrated HUDstats APIs to access stats and enrich content on its website as well. All information is aggregated in one place for the season. Casters can now easily access teams' and players' performance data throughout the season, creating a more engaging narrative. Teams have their own dedicated page with metrics and performance stats, offering greater visibility into their season achievements. They can share video highlights from the dashboard on social media channels, giving fans a front-row seat to their season-long journey.



Localizing content

Integrating Amazon Bedrock with their AVA solution, HUDstats provides their clients with a fully-streamlined solution of turning mere video feeds into content, be it video highlights or written social Media Content, match analysis or commentary feeds. The raw data extracted by HUDstats's AVA technology from the video feed is then fed into AWS's generative AI tools, which turn it into engaging content, at the push of a button. Users can even tweak their own prompts for advanced results. In this way, HUDstats enable their clients to generate an infinite number and types of content revolving around their competition, and in their own language.

Balanescu explained. "We have the whole suite of AWS generative AI tools at our disposal. Customers can download all their automatically-generated video highlights and post them immediately because they're all stored in S3. And once data is extracted from the video feeds by the computer vision algorithms and machine learning models, there are many incredible options to generate automated text and video highlights. We can say that Amazon Bedrock enables users with great creative freedom and inspiration, and we believe there's more to come."

"Moreover, AWS offers us the opportunity to optimize the costs and efficiency of the models because we can closely monitor how each model is performing in certain scenarios. As an example, using a model from a French company to generate content in French might produce better results as they might have a better and more improved French data set to train from. We are still learning, but our philosophy is looking at which model performs better for different use cases and apply that insight into our optimization. Amazon Bedrock offers us the best flexibility in this regard."

In addition to using Amazon Bedrock for generative AI, HUDstats leverages a full suite of AWS managed services. They're heavy users of Amazon Simple Storage Service ([Amazon S3](#)), Amazon Elastic Compute Cloud ([Amazon EC2](#)), and Amazon Elastic Kubernetes (Amazon [EKS](#)). They also use Amazon Managed Streaming for Apache Kafka ([MSK](#)), [ElastiCache](#), and Amazon Relational Database Service ([Amazon RDS](#)). Furthermore, HUDstats also leverage [CloudFront](#) for distribution of the front-end content, images and videos, [AWS Glue](#) for schemas and maintaining data integrity on the data warehouse, and [Amazon Athena](#) analytics service.

Advancing the state of the art for esports

For another project, the 2023 FIFAe Nations Cup esports tournament, HUDstats partnered with the [KNVB](#) (Royal Dutch Football Federation) to provide machine learning and AI technology to improve the streams of E_Oranje, the Dutch esports national football team. Their solution provided gameplay data and statistics such as shots on goal, passes, possession time, cards and penalties, enabling the team to create better player statistics, enhance storytelling capabilities and provide more insight into games.

For [DreamHack](#) Sports Games, HUDstack provided its AI-based AVA solution to fuel their tournament with in-depth match statistics, precise non-delayed data for live graphics, and deliver it with a staggering sub-second speed.

The company also created a user-friendly, fully-automated Competitive Gaming statistics platform for esports betting company Sports Information Services ([SIS](#)). The centralized hub for SIS' H2H Global Gaming League audience leverages AI technology to automatically capture and process data from Esoccer and Ebasketball gaming streams, turning them into insightful statistics.

Reflecting on the need for tools that keep pace with esports innovation, Balanescu said, "Having a mature setup like Amazon Bedrock is critical in bringing innovative ideas to clients. They've never seen this kind of thing before, so they can't make a list of requirements beforehand. We need to show them results in advance so they can react and understand the power of this technology. This is where AWS really shines. It allows us to do that conceptualization and iterating really fast, and gives us a mature setup that we can use to then expand and optimize."

AWS makes it easy, practical, and cost-effective for customers to use generative AI in their organizations. [Learn more about harnessing the power of generative AI with AWS.](#)